

CITS4401 Software Requirements and Design

Research Ethics Management System (REMS):

Phase 1 Requirements Engineering Report

Group 14

Team Members:

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1. Stakeholder Descriptions

Researcher:

Researchers are responsible for initiating and submitting research ethics applications and ensuring that their studies comply with ethical standards before involving human participants. They use the system to submit applications, upload supporting documents, and monitor the progress of their submissions. Compared to the current fragmented process, the proposed system provides a structured, centralised, and transparent workflow, enabling researchers to manage their applications more efficiently and reduce delays caused by manual communication methods.

Administrative Staff

Administrative staff are responsible for managing and processing research ethics applications. Their tasks include verifying application completeness, recording case notes, coordinating communication, and forwarding applications to the ethics committee.

Based on the interview conducted on 8 April 2026, it was identified that the current process relies heavily on manual methods such as spreadsheets and email communication. These practices lead to inefficiencies in tracking application status, managing document versions, and retrieving relevant files. The proposed system aims to centralise document management, provide real-time status tracking, and automate version control, thereby reducing manual workload, minimising errors, and improving overall operational efficiency.

Ethics Committee Member:

Ethics committee members are responsible for reviewing applications and making approval decisions. They use the system to access submitted materials, evaluate potential risks, and record decisions. The system supports them by providing organised information, a structured review process, and access to relevant application data, enabling more efficient and consistent decision-making.

These stakeholder roles were identified based on the REMS scenario and further refined through the interview conducted on 8 April 2026, which provided additional insights into the responsibilities and challenges faced by administrative staff in the current system.

2. User Stories

The following user stories are derived from the REMS scenario and the insights gained during the interview with administrative staff.

1. As a researcher, I want to submit an ethics application through a centralised system, so that I do not have to rely on fragmented tools such as email or shared platforms.
2. As a researcher, I want to upload all required supporting documents in one place, so that I can make sure my application is complete before submission.
3. As a researcher, I want to update or re-upload documents when requested, so that I can fix issues without restarting the entire application.
4. As an administrative staff member, I want all application documents to be stored in one centralised system, so that I do not need to search across emails, shared drives, and other platforms.
5. As an administrative staff member, I want to track the progress of applications throughout the review process, so that I am aware of the status even after forwarding them to the ethics committee.
6. As an administrative staff member, I want the system to maintain document version history automatically, so that I do not have to manually track versions using spreadsheets.
7. As an ethics committee member, I want to review applications and supporting documents within a structured system, so that I can make decisions more efficiently.
8. As an ethics committee member, I want access to a complete audit trail of actions, so that accountability and compliance can be ensured.

3. Interview Report

3.1 Interview Details:

The interview was conducted on 8 April 2026 via Microsoft Teams. The interviewee was an administrative staff representative involved in the management of research ethics applications. The session was attended by members of Group 14, who asked a series of questions to better understand the current workflow, document management practices, and system limitations. (One group member was unable to attend due to travel and limited network access; this was communicated to the interviewee in advance and acknowledged. The member subsequently reviewed the meeting recording and notes to ensure a full understanding of the discussion.)

3.2 Summary of Responses:

The interview provided a detailed overview of the current research ethics application process. Researchers are required to complete a questionnaire to determine a risk score, which defines the required supporting documents. Applications are then submitted through the system and reviewed by administrative staff for completeness before being forwarded to the ethics committee.

A major issue identified during the interview is the lack of effective document management. The current system only supports limited file formats, while other documents are managed through emails, shared drives, or external platforms. As a result, documents are often scattered across multiple locations, making them difficult to locate.

Document versioning is handled manually using spreadsheets. Administrative staff track file versions based on timestamps and file names, which was described in the interview as both time-consuming and error-prone. In addition, once an application is submitted, the system does not allow further file uploads, requiring all additional documents to be handled via email.

Another significant limitation identified is the lack of visibility in the review process. After forwarding applications to the ethics committee, administrative staff have no way of tracking progress and must rely solely on final notifications of approval or rejection.

The interview also highlighted that delays in the process are primarily caused by the need to locate and manage documents across multiple systems. Furthermore, it was noted that the

system lacks proper audit trail support, despite the requirement to maintain a complete record of actions for compliance purposes.

3.3 Key Insights and Impact:

The following summarizes key insights and their impact on requirements for each interview question:

Question 1: Current workflow from submission to final decision

Key Insight: The process uses a risk score to determine required documents; administrative staff check completeness before forwarding applications to the ethics committee.

Impact: Supports FR4 (application completeness verification).

Question 2: Document storage, versioning and management

Key Insight: Documents are scattered across email, SharePoint, and ERMS; version control is manual via Excel and highly error-prone.

Impact: Led to FR5 (centralised document repository) and FR6 (automatic version history).

Question 3: Limitations of current document upload

Key Insight: ERMS only allows one-time upload; post-submission updates must be handled via email.

Impact: Led to FR3 (post-submission document upload support).

Question 4: How application completeness is checked

Key Insight: Completeness is based on whether required documents match the risk category.

Impact: Supports FR4 (administrative review for completeness).

Question 5: How applications are assigned to ethics committee members

Key Insight: Assignments are based on risk level, with extreme risk cases handled by senior board members.

Impact: Informs stakeholder role understanding for FR9.

Question 6: How administrative staff track application status

Key Insight: Administrative staff have no visibility after forwarding applications, tracking relies on manual spreadsheets.

Impact: Led to FR8 (real-time status tracking).

Question 7: Main causes of delays and bottlenecks

Key Insight: Delays are primarily caused by searching for scattered documents and inconsistent file versions.

Impact: Reinforced the need for FR5 and FR6.

Question 8: What actions require audit tracking

Key Insight: All actions (submissions, edits, approvals, rejections) must be logged for at least five years for compliance.

Impact: Led to FR10 (audit trail) and NFR3.

Question 9: User roles and access permissions

Key Insight: Access is role-based; researchers see outcomes, administrative staff have limited access, and the ethics committee has full visibility.

Impact: Led to NFR2 (role-based access control).

The interview revealed several important limitations in the current system. One key issue is the heavy reliance on manual processes, such as spreadsheets and email communication, which makes the process inefficient and prone to errors. Another major limitation is the lack of a centralised document management system, resulting in documents being distributed across multiple platforms and making them difficult to locate, manage, and verify.

In addition, the absence of an integrated tracking mechanism prevents administrative staff from monitoring the progress of applications once they have been forwarded to the ethics committee, leading to reduced visibility and potential delays in the workflow.

These insights had a direct impact on the identification and refinement of system requirements. In particular, the need for centralised document storage, automated version control, and real-time application tracking functionality was identified as essential. The interview also highlighted the importance of maintaining a comprehensive audit trail to meet legal and compliance requirements. Overall, the interview provided valuable clarification of stakeholder needs and ensured that the proposed system design focuses on addressing the most critical inefficiencies and limitations in the current workflow.

4. Functional and Non-Functional Requirements

4.1 Functional Requirements [FR]:

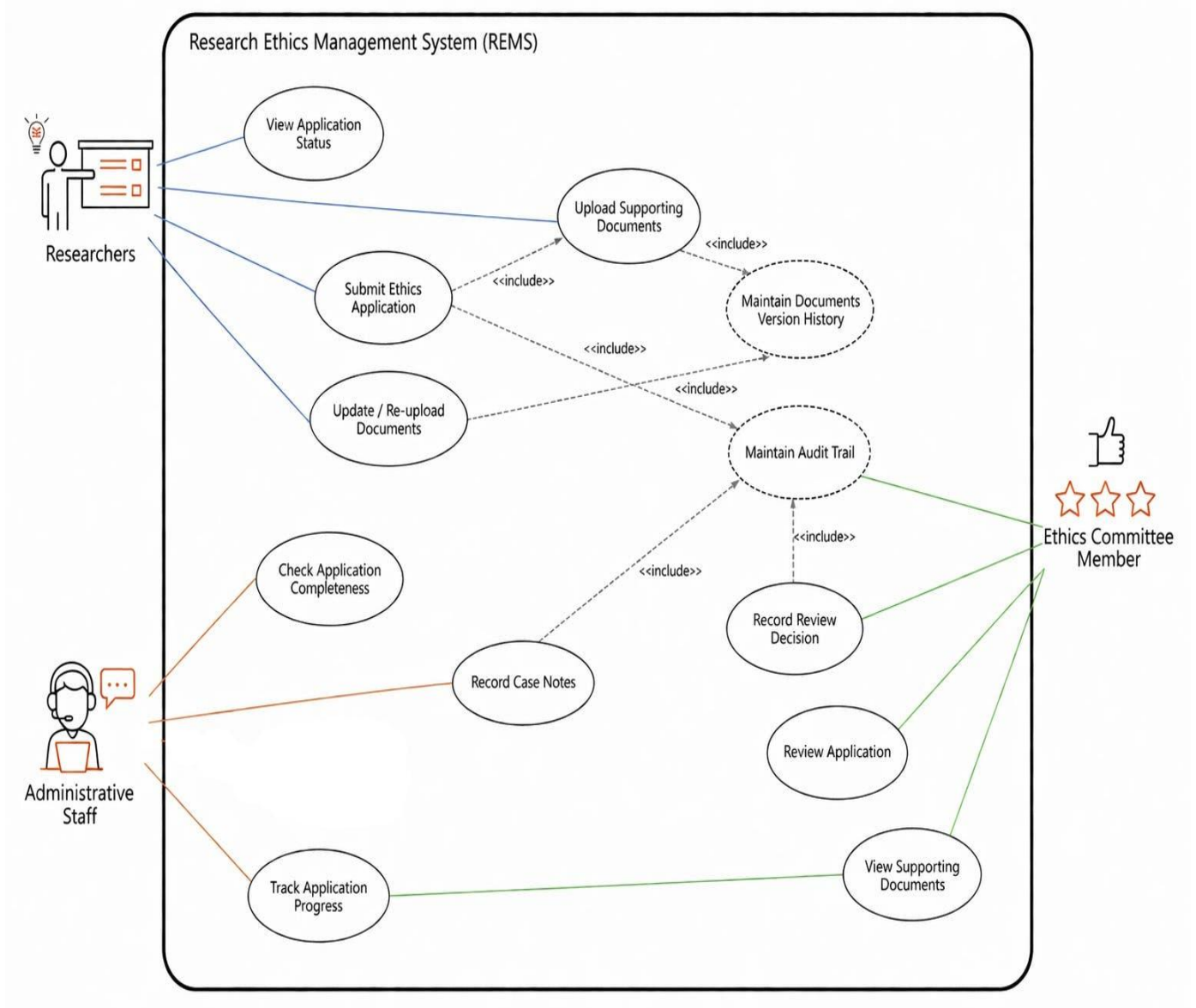
- **FR1:** The system shall allow researchers to submit research ethics applications through a centralised platform.
Source: Interview finding (fragmented submission tools)
- **FR2:** The system shall allow researchers to upload all required supporting documents during application submission.
Source: Interview finding (scattered documents)
- **FR3:** The system shall allow researchers to upload additional or updated documents after submission when requested by administrative staff.
Source: Interview finding (no post-submission document support)
- **FR4:** The system shall allow administrative staff to review applications and verify completeness based on required documents.
Source: Stakeholder role (administrative staff responsibility)
- **FR5:** The system shall store all application-related documents in a centralised repository accessible to authorised users.
Source: Interview finding (fragmented document storage)
- **FR6:** The system shall maintain version history for all uploaded documents, including timestamps and previous versions.
Source: Interview finding (manual version tracking)
- **FR7:** The system shall allow administrative staff to record, edit, and view case notes associated with each application.
Source: Stakeholder role (administrative staff workflow)

- **FR8:** The system shall display the current status of each application (e.g., Submitted, Under Review, Approved, Rejected) to authorised users.
Source: Interview finding (lack of process visibility)
- **FR9:** The system shall allow ethics committee members to access, review, and evaluate submitted applications and supporting documents.
Source: Stakeholder role (ethics committee review duty)
- **FR10:** The system shall record all system actions (including submissions, updates, reviews, and decisions) in an audit trail.
Source: Interview finding (missing audit trail for compliance)

4.2 Non-functional Requirements [NFR]:

- **NFR1:** The system shall ensure that all sensitive data is securely stored using encryption mechanisms and protected from unauthorised access.
- **NFR2:** The system shall enforce role-based access control to ensure users can only access information relevant to their assigned role.
- **NFR3:** The system shall maintain a complete audit trail of all system actions for a minimum of five years to support legal and compliance requirements.
- **NFR4:** The system shall provide an intuitive user interface that allows users to complete key tasks (e.g., application submission) within 5 minutes under normal conditions.
- **NFR5:** The system shall allow users to retrieve documents within 2 seconds under normal operating conditions.
- **NFR6:** The system shall ensure data reliability by preventing data loss during normal system operation and maintaining data integrity.
- **NFR7:** The system shall achieve at least 99% availability, excluding scheduled maintenance periods.
- **NFR8:** The system shall support concurrent access by multiple users without performance degradation under normal usage conditions.

5. Use Case Diagram



6. Detailed Use Case Description

This section presents a detailed use case describing how a researcher submits a research ethics application through the system.

Use Case Name: Submit Research Ethics Application

Primary Actor: Researcher

Goal:

The goal of this use case is to allow a researcher to submit a complete research ethics application, along with all the required supporting documents, so that it can be reviewed by administrative staff.

Preconditions:

- The researcher has access to the system and is logged in.
- The researcher has already completed the risk assessment questionnaire.
- The researcher has all the necessary supporting documents ready.

Main Success Scenario:

1. The researcher logs into the system.
2. The researcher chooses to start a new ethics application.
3. The system displays a form asking for all the required details.
4. The researcher fills in the application information, including the risk score.
5. Based on the risk category, the system shows a list of required supporting documents.
6. The researcher uploads all the required documents.
7. The system checks whether all required fields and documents have been provided.
8. The researcher reviews the entered information and uploaded documents.
9. The researcher submits the application.
10. The system saves the application and all associated documents in the system.

11. The system records the submission in the audit trail.
12. The system updates the application status to “Submitted.”
13. The application is now available for administrative staff to review.

Alternative / Exceptional Flows:

A1. Missing information:

If some required details are not filled in, the system shows an error message and does not allow submission until everything is completed.

A2. Missing documents:

If the required documents are not uploaded, the system notifies the researcher and prevents submission until all documents are provided.

A3. Invalid file format:

If a file is uploaded in an unsupported format, the system rejects it and asks the researcher to upload a valid file.

A4. System error during submission:

If something goes wrong during submission, the system stops the process and informs the researcher that the submission has failed.

A5. Duplicate submission:

If the researcher tries to submit the same application again, the system detects it and prevents duplication, informing the user accordingly.

Postconditions:

- The application is successfully stored in the system.
- All supporting documents are linked to the application.
- The submission is recorded in the audit trail.
- The application status is set to “Submitted.”
- Administrative staff can access the application for completeness checking.